

RESEARCH ARTICLE

Adaptive Asynchronous Split Federated Learning for Medical Image Segmentation

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ABSTRACT Split federated (SplitFed) learning offers promise for collaborative machine learning across decentralized and resource-constrained clients (edge devices, nodes, or organizations) in various applications, including healthcare. However, real-world challenges arise in heterogeneous environments where clients experience communication-related data losses, differ in computational capabilities, and have varying local dataset sizes. In this paper, we introduce adaptive asynchronous split federated learning (AASFL), an innovative training scheme to address such challenges. AASFL blends asynchronous SplitFed learning with client-level adaptability, acknowledging varying client capabilities. In AASFL, each client independently adapts its learning rate and the number of local training epochs based on its local training speed, dataset size, and packet loss probability. To demonstrate the effectiveness of AASFL, we implement and evaluate it on a SplitFed network architecture for medical image segmentation. Moreover, we demonstrate the validity of AASFL by emphasizing the necessity of training each client in the collaborative network and revealing drawbacks of client selection. The results on two public datasets indicate that it greatly enhances global model performance, leading to more accurate segmentation results. We also statistically show that AASFL outperforms the standard SplitFed. Furthermore, we provide a theoretical convergence analysis of AASFL. To the best of our knowledge, this is the first analysis of SplitFed in an adaptive, asynchronous setting. The proposed AASFL training scheme offers a promising avenue for improving medical image segmentation tasks in practical decentralized learning environments.

INDEX TERMS Adaptive training, asynchronous training, medical image segmentation, split federated learning.

I. INTRODUCTION

Medical image segmentation plays a pivotal role in healthcare research. This critical task is central to the diagnosis, treatment planning, and monitoring of various medical conditions. Deep neural networks (DNNs) have been used increasingly in medical image segmentation in recent decades [1], [2], [3], [4]. As the volume of medical image data continues to grow, the demand for accurate and efficient medical image segmentation has also been growing rapidly. Stakeholders

who may benefit from medical image segmentation, who also contribute data for model training, include individual patients, healthcare providers, hospitals, laboratories, medical researchers, etc. Centrally training DNN models using clients' data is often infeasible due to privacy concerns, data ownership issues, and healthcare-specific laws and regulations.^{1,2} Therefore, training such models in a decentralized, collaborative manner is necessary. Decentralized learning techniques such as Federated Learning (FL) [5], Split

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¹<https://www.osano.com/articles/data-privacy-laws>

²<https://www.priv.gc.ca/en/privacy-topics/privacy-laws-in-canada>

Learning (SL) [6] and Split Federated Learning (SplitFed) [7] facilitate collaborative model training across multiple clients, offering solutions for these challenges [8].

FL addresses privacy concerns through collaborative model training across a distributed network of clients [5]. SL distributes model parts to multiple locations to overcome resource constraints. In SL, the model is divided at split points, with a larger portion on the server side and a smaller portion on the client side. SL also enhances data privacy by storing data and labels in separate locations [6]. SplitFed combines the strengths of FL and SL, allowing clients with limited computational capabilities to participate in collaborative model training while preserving data privacy [7].

In collaborative networks, clients and servers communicate with each other synchronously or asynchronously. Conventional federated and SplitFed learning approaches mostly rely on synchronized communication, where each client sends updates to a central server for model aggregation in a coordinated manner [5], [7], [9]. Synchronous approaches are effective in improving global model performance and preserving data privacy, but face challenges with client heterogeneity. In FL, clients can broadly be divided into two types: leaders (faster-processing clients) and stragglers (slow clients) [10], [11]. Obviously, stragglers can cause delays in the federated learning process. Asynchronous communication addresses these challenges by allowing clients to update the model at their own pace, improving flexibility and error handling in decentralized learning [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22].

In healthcare collaborations, client diversity and privacy preservation are two main concerns. Therefore, the traditional approach of treating all clients in the same way poses certain risks. Adaptive training addresses this challenge by adjusting the learning processes for each client during local training, thereby enhancing the ability of each client to contribute to the collaborative effort in the best way consistent with its own constraints [20], [23], [24], [25].

In this paper, we present an adaptive and asynchronous training scheme to address the above-mentioned challenges in SplitFed learning. The contributions of the paper are summarized as follows:

- We introduce adaptive asynchronous split federated learning (AASFL) as a novel training scheme for split federated medical image segmentation. AASFL combines the benefits of asynchronous training with a unique form of adaptability at the client level.
- We classify clients into two groups based on the packet loss probability (P_L) in their communication channels. Using the proposed AASFL training scheme, these two groups may be treated differently during the training process by performing parameter aggregation for one group more frequently than the other. This is where we employ asynchronicity.
- In our proposed AASFL training scheme, each client adaptively decides its learning rate and the number

of local training epochs, based on its local processing speed, network reliability, data volume, and packet loss.

- We highlight the importance of preventing clients from being excluded from collaborative training via client selection. Although state-of-the-art (SOTA) decentralized research shows that client selection improves overall performance [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], our experimental results indicate that client selection might not always be beneficial, especially when adaptive and asynchronous methods are employed.
- Our experimental results on two medical image segmentation datasets demonstrate that AASFL outperforms five well-known parameter aggregation methods for SplitFed learning.
- Finally, we provide a theoretical convergence analysis for the AASFL training scheme. To the best of our knowledge, this is the first convergence analysis of split federated learning in an adaptive and asynchronous training environment.

The rest of this paper is organized as follows: We first discuss related works in Section II. In Section III, we present the proposed AASFL training scheme. Section V provides the convergence analysis of the AASFL. Our experimental results are then presented in Section IV. Finally, we present the conclusions and suggestions for future work in Section V.

II. RELATED WORKS

This section outlines the related works associated with communication challenges, adaptive training, and asynchronous training in federated and SplitFed networks, followed by details on client selection. While adaptive and asynchronous techniques have been predominantly used in the context of FL, their utilization in SplitFed is relatively new.

A. COMMUNICATION CHALLENGES IN FEDERATED AND SPLITFED LEARNING

Communication is one of the key aspects of decentralized learning. Some of the associated challenges are communication overhead, network latency, bandwidth constraints, performance degradations caused by stragglers, dynamic network conditions, asynchronous updates, etc. Mitigating communication challenges in FL was extensively studied in previous research [8], [36], [37], [38], [39], [40], [41], [42]. More focused research was on developing robust models in the presence of noisy channels [43], fading channels [44], and packet loss [45], [46], [47]. Examples of related research on SplitFed learning are [48] and [49], which deal with noisy channels and packet loss, respectively.

B. ADAPTIVE TRAINING IN FEDERATED NETWORKS

The first work on adaptive training of decentralized networks was proposed in [23], showing the importance of adaptive learning rates. Another study reported in [24] showed that the time cost of a set of heterogeneous clients' local training

reflects their training speed and can be used to guide the dynamic setting of local epochs. The adjusting mini-batch and local epoch (AMBLE) method proposed in [20] is another adaptive algorithm, in which the local mini batch size, local epochs and learning rates were adaptively decided for each client. Another recent study presented in [25] proposes a scaling rule to determine the learning rate based on the batch size.

C. ASYNCHRONOUS TRAINING IN FEDERATED AND SPLITFED NETWORKS

Digital twin (DT)-enabled asynchronous SplitFed learning (DT-ASFL) [12] was the first recorded SplitFed system to improve communication in an e-healthcare system by letting client updates to the global model happen at different times. FedSA [13] is a semi-asynchronous FL mechanism developed for heterogeneous edge computing. In FedSA, the parameter server only updates a certain number of local models based on their arrival order. FedBuff [14] is an asynchronous federated optimization framework that uses buffered asynchronous aggregation. FedBuff's server aggregates only a subset of client updates in a secure buffer before performing a server update. AsyPFL [15] mitigates clients' erratic behaviour by adding an elastic term to their local objective. Their approach became advantageous by allowing asynchronous updates from multiple clients and being robust against high communication delays with an asynchrony-related parameter. PersA-FL [16] considered two optimization-based frameworks: (i) Model-Agnostic Meta-Learning (MAML) and (ii) Moreau Envelope (ME) for personalization. They removed synchronous communication to improve the scalability of their networks. TimelyFL [17] is a recent asynchronous FL method that adjusts the local training environment based on real-time resources of the clients, which facilitates more clients joining collaborative training without staleness.

D. CLIENT SELECTION

1) THE IMPORTANCE OF CLIENT SELECTION

Client selection has proven to be an important consideration in the FL literature, impacting model quality, communication efficiency, privacy, system robustness, etc. [26], [27], [28], [29], [30], [31], [32], [33], [34], [35]. It refers to the process of choosing clients to participate in a particular model training round. When the number of clients connecting in the collaborative network increases, the overhead of managing all the clients slows down the overall training process. Therefore, client selection benefits collaborative networks with an overwhelming number of clients. It is reported in [5] that the global model performance shows diminishing returns when clients are added beyond a certain point. Their logic was that if the algorithm chose to include every client as opposed to only a subset, it would negatively affect the global model performance while increasing the communication cost.

Various strategies exist for client selection [27], [28], [29], [30], [31], [32], [33], [34], [35], including: (i) *Random selection*: clients are chosen randomly [35], (ii) *Stratified sampling*: clients from different subgroups of the population are represented [33], [34], (iii) *Performance-based selection*: clients with better performance, high-quality data, or superior computational capabilities are chosen [28], (iv) *Proximity-based selection*: clients closer to the central server are chosen [31], (v) *Privacy-preserving selection*: clients that maintain overall privacy of the network are chosen [28], (vi) *Dynamic/adaptive selection*: clients with suitable adaptability in terms of connectivity, computational power, and recent model update history are chosen [28], and (vii) *Strategic sampling*: clients with unique data or values are chosen [28], [30].

2) LIMITATIONS OF CLIENT SELECTION

Even though client selection holds the potential to mitigate associated challenges, it may not always be straightforward or beneficial in a collaborative learning environment [50], [51]. Firstly, when it comes to the heterogeneous aspects of clients, selecting clients indiscriminately may lead to the absence of a heterogeneous set of participants. Therefore, training data on the selected clients may not adequately represent the overall data distribution and may impact the generalization of the global model. Secondly, preserving privacy is crucial in collaborative networks. Improper client selection can compromise sensitive information. Therefore, careful client selection is essential to maintain proper security standards in the network. Thirdly, client selection would impact the fairness of the global model. Such an indiscriminate client selection may lead to biased representations, favouring a set of clients over others. In the meantime, client selection is problematic when operating in dynamic environments where clients can often join or leave the network, especially in collaborative networks in healthcare. Therefore, client selection, especially during the early stages of global model training, would not always be beneficial.

The proposed AASFL training scheme builds upon SOTA-FL and SplitFed research. Similar to these related works, AASFL aims to improve the global model's performance. However, it sets itself apart by comprehensively integrating adaptability and asynchronicity into emerging SplitFed networks. Existing works on adaptability are based upon factors such as processing speed, network reliability, and data volume. AASFL goes a step further by also addressing packet loss, a critical concern in healthcare collaborations. Likewise, existing asynchronous training methods often rely on partial clients' participation during model aggregation. AASFL incorporates asynchronicity via full clients' participation by treating the identified two client groups differently during training. Furthermore, while existing works prioritize client selection to improve performance, AASFL demonstrates that it may not always be beneficial, especially in adaptive and asynchronous training scenarios. Overall, the proposed AASFL training scheme innovatively

combines adaptability and asynchronicity to ensure full client participation in split federated medical image segmentation.

III. ADAPTIVE ASYNCHRONOUS SPLIT FEDERATED LEARNING (AASFL)

In this section, we present our proposed AASFL training scheme. First, we describe some preliminary notation that is used in the SplitFed training process and the optimization objectives of the AASFL training scheme. We then describe the AASFL training scheme in detail.

A. PRELIMINARIES

Consider a set of clients $\mathcal{N} = \{1, 2, \dots, N\}$. Each client $n \in \mathcal{N}$ contains its own local dataset $\mathcal{D}_n$ of size $|\mathcal{D}_n|$. Each data sample in $\mathcal{D}_n$ consists of an image and its corresponding ground-truth (GT) data. Using each client's unique local dataset, the clients in $\mathcal{N}$ jointly train a deep learning model, denoted by w , whose goal is to map an image to its corresponding GT. In this paper, we assume that all clients use the same architecture. Each client model is split at two points that create three separate model parts, which are termed the frontend (FE) model, w_{FE} , server (S) model (that is resided in the main server), w_S , and backend (BE) model, w_{BE} . Therefore, $w := [w_{FE}; w_S; w_{BE}]$. As previously mentioned, model splitting is done as a solution for the resource constraint problem with clients and to preserve the privacy of images and their GTs.

1) SPLITFED TRAINING PROCESS

Our SplitFed architecture, depicted in Fig. 1, follows the design proposed in [7]. The federated server occupies the FE model (w_{FE}), which is the global FE model, and the BE model (w_{BE}), which is the global BE model. The main server occupies the S model (w_S), which is the global S model. Federated server sends initial copies of the FE and BE models to each client. The main server makes a separate copy of its own model for each client. Each client then locally trains its FE model and BE model in collaboration with its own copy of the S model for a certain number of local epochs (E) at some learning rate (η). This is called the **local update**. Each client performs **local validation** after each epoch and records its local validation loss (L_n) after its final epoch. After the local update and validation, each client sends its locally-trained FE model and BE model to the server, which is known as the **push update**. Then, locally-trained model updates of each client's FE, S, and BE models are combined together, which is known as **parameter aggregation**. Parameter aggregation of the FE and BE models happens at the federated server, while parameter aggregation of the S models happens at the main server. The combined model is the new global model, which consists of the updated FE, the S, and the BE models. Therefore, these three models are the updated global FE, global S, and global BE models. This completes one **communication round (com round)**. One such parameter aggregation is referred to as a **global update**. Therefore, after each com round, one global update happens. The server

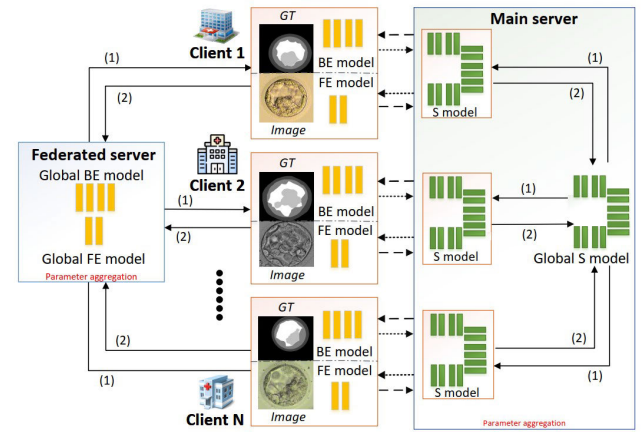


FIGURE 1. SplitFed architecture with multiple clients, where each client uses the split U-Net. FE and BE stand for the frontend and the backend, respectively, and S stands for the server side. Marked with (1) are the transmissions of the current FE and BE global models from the federated server to the clients and the transmissions of the current S global model from the main server to the clients' copies of the server model. Marked with (2) are the transmissions of the locally-trained FE and BE local models from the clients to the federated server and transmissions of the locally-trained S copies of local models from the server model to the main server. The arrows with longer dashes indicate the features flow, and the arrows with shorter dashes indicate the error gradient flow. GT refers to ground-truth.

sends the global FE and the global BE models to each client to perform **global validation**. The global FE, global S, and global BE models are then used to perform the **global testing** after each com round.

2) OPTIMIZATION GOALS OF THE AASFL TRAINING SCHEME

This sub-section defines the optimization goals of AASFL. First, we introduce the following novel terminologies that make the AASFL training process different from the standard SplitFed:

- R clients refer to “Reliable” clients that have clean or perfect communication channels at their model split points. C_R is the set of R clients.
- NR clients refer to “Non-Reliable” clients that have communication channels with packet loss at their model split points. C_{NR} is the set of NR clients.
- **R-com round** refers to a com round of only R clients.
- **T-com round** refers to a com round of all the clients (both R and NR). T refers to *true* to be in line with the standard SplitFed approach.
- **R-global update** refers to the parameter aggregation of R clients after the completion of each R -com round.
- **T-global update** refers to the parameter aggregation of all the clients (both R and NR) after the completion of each T -com round.
- T_m^* , $m \in \{1, 2, \dots, M\}$ stands for the timesteps when T -com rounds are completed or T -global updates are done. M is the number of T -com rounds.
- $t_{m,k}$, $m \in \{1, 2, \dots, M\}$ and $k \in \{1, 2, \dots, K\}$ stands for the timesteps when R -com rounds are completed or

R -global updates are done. K is the number of R -com rounds for each T -com round.

Consider that model w is obtained through the local updates of clients. Let $f_n(w)$ be the training loss of model w of the n^{th} client. Let $f_{R,n}(w) := \mathbb{E}_{d_i \in D_{R,n}}[f_n(w)]$ represents the expected loss of model w over the n^{th} R client's local dataset, where $D_{R,n}$ is the size of the n^{th} R client's local dataset. The goal of the R -global update is to minimize the expected loss of the model w over the local datasets of all the R clients, $F_R(w)$, which can be expressed as:

$$\min_w F_R(w) := \sum_{n \in C_R} p_{R,n} f_{R,n}(w), \quad (1)$$

where $p_{R,n}$ is the normalized data ratio of the n^{th} R client. We typically have:

$$p_{R,n} = \frac{D_{R,n}}{\sum_{n \in C_R} D_{R,n}}, \quad (2)$$

such that $p_{R,n} \geq 0$ and $\sum_{n \in C_R} p_{R,n} = 1$.

The goal of the T -global update is to minimize the loss $F(w)$ of the model w over the datasets of all the clients (both R and NR), which can be expressed as:

$$\min_w F(w) := \min_w \sum_{n=1}^N l_n f_n(w), \quad (3)$$

where l_n is the local validation loss ratio of the n^{th} client:

$$l_n = \frac{\frac{1}{L_n}}{\sum_{n=1}^N \frac{1}{L_n}}, \quad (4)$$

such that $l_n \geq 0$ and $\sum_{n=1}^N l_n = 1$.

B. PROPOSED AASFL TRAINING SCHEME

Before presenting the proposed AASFL methodology, we first discuss the insight about the optimal choice of the local number of epochs E_n and the learning rate η_n for client n . This insight is obtained from the convergence analysis in the Appendix. In Theorem 1 in the Appendix, the upper bounds on the expected deviations of FE, BE, and S models from the optimal model for client n are expressed in terms of E_n and η_n . As an example, we show the bound for the S model of client n (the bounds for FE and BE models are analogous):

$$\mathbb{E}[\|w_S^{E_n} - w_S^*\|^2] \leq (1 - \mu\eta_n)^{E_n} \|w_S^0 - w_S^*\|^2 + \frac{2\eta_n^2 r^2 E_n}{N^2}, \quad (5)$$

where w_S^* are the optimal weights for the S model, w_S^0 are the initial weights, $w_S^{E_n}$ are the weights after E_n local epochs, μ is the constant of μ -strong convexity (assumption 2 in the Appendix), r^2 is the bound on the stochastic gradient square norm (assumption 3 in the Appendix), and N is the total number of clients. The goal is to select E_n and η_n to minimize the upper bound on the right-hand side of (5), so that the expected deviation from the optimal model is minimized.

Algorithm 1 Adaptive Asynchronous Split Federated Learning

Input: N, C_R, C_{NR}, K, M

Output: $w^M = [w_{T,FE}^M; w_{T,S}^M; w_{T,BE}^M]$

Initialization: $w^0 = [w_{T,FE}^0; w_{T,S}^0; w_{T,BE}^0]$

```

1: Step 1: Screening communication (com) round
   one_training_epoch( $C_R \cup C_{NR}$ ) with all clients
   Set  $E_n$  and  $\eta_n$  as in (14) and (15)
2: for  $m = 1, 2, \dots, M$  do
3:   Step 2: R clients' local update
     one_training_epoch( $C_R$ ) with  $R$  clients
     Iterate through calculated  $E_n$  with calculated  $\eta_n$ 
4:   for  $j = 1, 2, \dots, K$  do
5:     Step 3: R-global update
       aggregation of  $R$  clients
       Update  $w^{m,Rj} = [w_{R,FE}^j; w_{R,S}^j; w_{R,BE}^j]$ 
6:   end for
7:   Step 4: NR clients' local update
     one_training_epoch( $C_{NR}$ ) with  $NR$  clients
     Iterate through calculated  $E_n$  with calculated  $\eta_n$ 
     Parameter aggregation of  $NR$  clients
8:   Step 5: T-global update
     Parameter aggregation of all the clients
     Update  $w^m = [w_{T,FE}^m; w_{T,S}^m; w_{T,BE}^m]$ 
9: end for
10: one_training_epoch ( $C$ ):
11: for each client  $c \in C$  do
12:   # Forward pass in parallel
13:   FE forward update: forward pass in the FE model
     FE model sends features to the S model
14:   S forward update: forward pass in the S model
     S model sends features to the BE model
15:   BE forward update: forward pass in the BE model
16:   # Forward pass completes
17:   BE model computes error gradients
18:   # Backward pass in parallel
19:   BE backward update: Update  $w_{BE}$ 
     BE model sends gradients to the S model
20:   S backward update: Update  $w_S$ 
     S model sends gradients to the FE model
21:   FE backward update: Update  $w_{FE}$ 
22:   # Backward pass completes
23: end for

```

First, note that $\mu\eta_n > 0$, since both values are positive. Also, for the bound to be non-negative for all choices of E_n , we should have $\mu\eta_n < 1$. Expanding $(1 - \mu\eta_n)^{E_n}$:

$$(1 - \mu\eta_n)^{E_n} = 1 - E_n\mu\eta_n + O((\mu\eta_n)^2), \quad (6)$$

and keeping only the first two terms,

$$(1 - \mu\eta_n)^{E_n} \approx 1 - E_n\mu\eta_n. \quad (7)$$

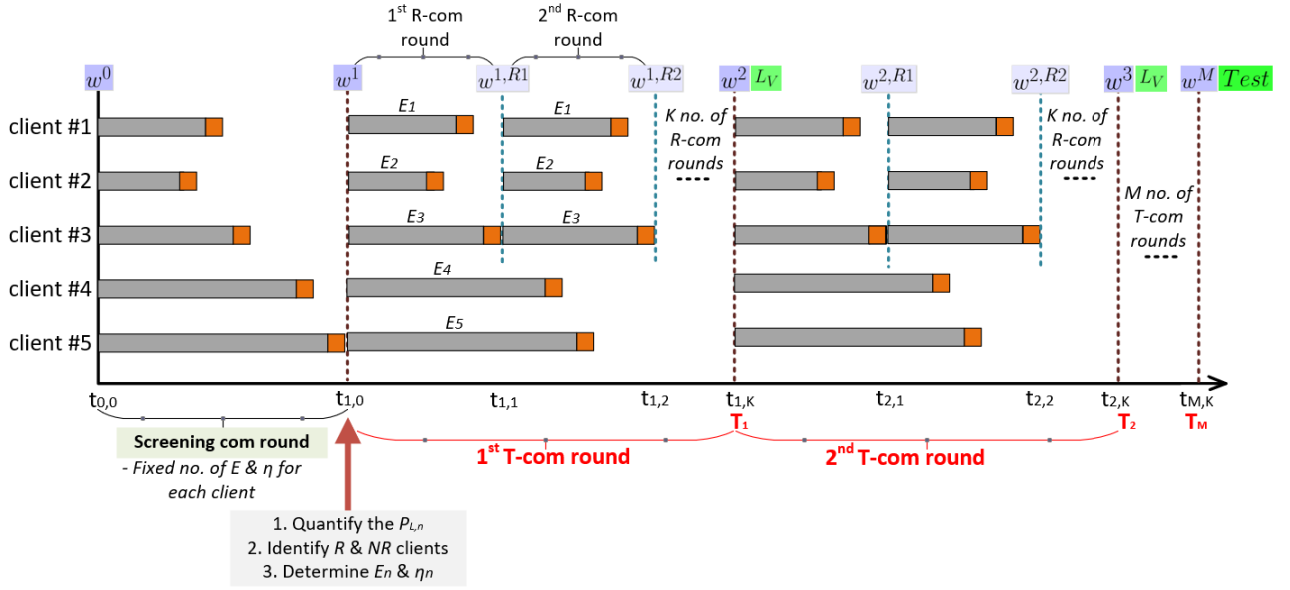


FIGURE 2. An example of SplitFed network with the AASFL training scheme using five clients. The ash-colored segments represent local updates from clients. The orange-colored segments represent the push updates. Blue dotted lines indicate R -global updates. Brown dotted lines indicate T -global updates. w^0 represents the initial global model. $w^{M,RK}$ is the R -global model created at the K^{th} R -com round of the M^{th} T -com round. w^M is the T -global model at the M^{th} T -com round. L_V stands for local validation on the T -global models. Test stands for testing. $T_m, m \in \{1, 2, \dots, M\}$ stands for the timesteps when T -com rounds are completed or T -global updates are done. M is the number of T -com rounds. $t_{m,k}, m \in \{1, 2, \dots, M\}$ and $k \in \{0, 1, \dots, K\}$ stand for the timesteps when R -com rounds are completed or R -global updates are done. K is the number of R -com rounds for each T -com round.

Using this approximation, the bound in (5) becomes

$$(1 - E_n \mu \eta_n) \cdot \|w_S^0 - w_S^*\|^2 + \frac{2\eta_n^2 r^2 E_n}{N^2}. \quad (8)$$

This is a quadratic expression in η_n with a positive quadratic coefficient, hence it has a minimum. Differentiating with respect to η_n and setting the derivative to zero, we obtain the optimal learning rate as:

$$\eta_n = \frac{1}{4r^2} \mu \cdot \|w_S^0 - w_S^*\|^2 \cdot N^2. \quad (9)$$

Note that μ , r^2 , and w_S^* are unknown, so we cannot use (9) directly to compute η_n , but can use the insight that the optimal learning rate is proportional to N^2 :

$$\eta_n \propto N^2. \quad (10)$$

To derive similar insight for E_n , we can use (9) in (8) and after some algebraic rearrangement obtain the following expression for the bound:

$$\|w_S^0 - w_S^*\|^2 - E_n \cdot \frac{1}{8r^2} \cdot \mu^2 \|w_S^0 - w_S^*\|^4 \cdot N^2. \quad (11)$$

In order to minimize this bound, we want to pick E_n to be as large as possible while keeping the bound non-negative. It follows that E_n should be the largest integer such that:

$$E_n \leq \frac{8r^2}{\mu^2 \cdot \|w_S^0 - w_S^*\|^2 \cdot N^2}. \quad (12)$$

Again, we cannot use this expression directly to compute E_n since it contains unknown constants, but the insight is that the optimal E_n is proportional to $1/N^2$:

$$E_n \propto \frac{1}{N^2}. \quad (13)$$

These insights are used in the proposed AASFL to set the values for E_n and η_n .

The proposed AASFL training scheme is summarized in Algorithm 1. It has five steps: **Step 1: Screening communication (com) round** (line 1), **Step 2: R clients' local update** (line 3), **Step 3: R -global update** (lines 4-6), **Step 4: NR clients' local update** (line 7), and **Step 5: T -global update** (line 8). The inputs to the training scheme are N , K , and M . Output is the global model w^M created in the M^{th} T -com round, where $w^M = [w_{T,FE}^M; w_{T,S}^M; w_{T,BE}^M]$. The initial global model is represented as w^0 , where $w^0 = [w_{T,FE}^0; w_{T,S}^0; w_{T,BE}^0]$. The function `one_training_epoch(C)` implements one forward and backward pass of any client $c \in C$ through FE, S, and BE models (lines 10-23). C is the set of clients, which could be C_R , C_{NR} , or their union (all clients). A more detailed description of each of these steps is given below. Please also refer to Fig. 2, which shows a SplitFed network with the AASFL training scheme, along with this description.

Step 1: Screening communication (com) round: The screening com round is performed to obtain the characteristics of the clients participating in SplitFed training. It involves screening every client prior to the actual training process. In Fig. 2, $t_{0,0}$ is the starting time of the AASFL process, and

the screening com round occurs in the time interval $[t_{0,0}, t_{1,0}]$. A copy of the initial global model w^0 is distributed to all participating clients. Each client trains its own copy of the model parts for one local epoch at a fixed learning rate. This completes the screening com round. Here, we set the learning rate to 1×10^{-4} , similar to the learning rate we used in the centralized and standard SplitFed settings in [48], [49]. After the screening com round, the following actions will take place:

- P_L is computed for each client n ($P_{L,n}$).
- Clients are divided into reliable and non-reliable ones based on $P_{L,n}$. If $P_{L,n} > 0$, client n is assumed to be a reliable client (R) and non-reliable (NR) otherwise.
- The number of local epochs and the learning rate for each client n is determined as discussed below.

Following (13), the number of local epochs for client n is set as:

$$E_n = \text{round} \left(\left[\alpha \frac{T_{slowest}}{T_n^{screen}} + \beta \frac{D_{max}}{D_n} + \gamma(1 - P_{L,n}) \right] \cdot \frac{1}{N^2} \right), \quad (14)$$

where $T_{slowest}$ is the maximum amount of time a client requires to complete one local epoch during the screening com round (making it the *slowest client*), T_n^{screen} is the time taken by the n^{th} client to complete one local epoch during the screening com round, and D_{max} is the size of the largest training dataset of the participating clients. α , β , and γ are the parameters to tune such that E_n values are in between a pre-defined range ($8 \leq E_n \leq 12$). This range was experimentally found for the two datasets we used in our experiments. In Section III-C, we will provide a more detailed rationale for each of the terms in (14).

Following (10), the learning rate for client n is set as:

$$\eta_n = \left[\delta \frac{\frac{D_{max}}{D_n}}{\frac{T_{slowest}}{T_n^{screen}}} + \sigma(1 - P_{L,n}) \right] \cdot N^2, \quad (15)$$

where δ and σ are the parameters to tune such that η_n values are in between a pre-defined range ($1 \times 10^{-5} \leq \eta_n \leq 1 \times 10^{-4}$). This range was experimentally decided for the two datasets that we used in our experiments. In Section III-C, we will present the rationale behind each term in (15).

After the screening com round, parameter aggregation of all the clients happens based on the client's normalized data ratio, p_n , which is defined by:

$$p_n = \frac{D_n}{\sum_{n=1}^N D_n}, \quad (16)$$

such that $p_n \geq 0$ and $\sum_{n=1}^N p_n = 1$. A new global model (w^1) is made in the timestep $t_{1,0}$. A copy of w^1 is then received by each client.

Step 2: R clients' local update: Each R client trains their copy of the w^1 adaptively for the calculated number of E_n with the calculated η_n . It is assumed that the learning rate is constant over the epochs and also the com rounds. In Fig. 2,

R clients' 1^{st} local update during the 1^{st} T -com round occurs during the $[t_{1,0} - t_{1,1}]$ time period.

Step 3: R -global update: When the group of R clients finishes the local update, an R -global update happens. Parameter aggregation with R clients happens based on R clients' normalized data ratio ($p_{R,n}$), using equation (2). A new R -global model ($w^{1,R1}$) is made at the timestep $t_{1,2}$. Copies of $w^{1,R1}$ are distributed to R clients, and local training is continued. The R -global model created at the K^{th} R -com round of the M^{th} T -com round is referred to as $w^{M,RK}$. Here, we used $K = 3$ for the trade-off between performance and complexity.

Step 4: NR clients' local update: As discussed earlier, during the screening com round, we identified a group of NR clients. When R clients are completing more R -com rounds, NR clients may still continue with their local updates, since they have a low learning rate or may be waiting until they receive the updated global model for the next round of local updates. In this waiting period, NR clients can make the decision about re-transmission or packet loss correction methods if they have the capabilities or resources to do so. Parameter aggregation with NR clients happens based on NR clients' modified data ratio, $p_{NR,n}$, which is defined by:

$$p_{NR,n} = \frac{e^{D_{NR,n}}}{\sum_{n \in C_{NR}} e^{D_{NR,n}}}, \quad (17)$$

such that $p_{NR,n} \geq 0$, and $\sum_{n \in C_{NR}} p_{NR,n} = 1$. $D_{NR,n}$ is the size of the n^{th} NR client's local dataset. We used the exponential term in the data ratios of the NR clients as a normalization strategy. In Fig. 2, NR clients' local update during the 1^{st} T -com round occurs during the $[t_{1,0} - T_1]$ time period.

Step 5: T -global update: When all clients complete their local updates, a T -global update happens. Parameter aggregation is based on all client's local validation loss ratio (l_n), using equation (4). A new T -global model (w^2) is made at the timestep T_1 . Local validation on the T -global model is performed (L_V).

We repeated this process M times. The T -global model created at the M^{th} T -com round is referred to as w^M . We used $M = 10$ for our experiments, similar to the standard SplitFed settings in [48] and [49]. Finally, testing ($Test$) is performed on w^{10} using the global test test. Mean Jaccard Index (MJJI) [52] is recorded.

C. THE RATIONALE BEHIND E_n AND η_n

In this section, we explain the rationale behind expressions for E_n and η_n in (14) and (15), respectively. First, note that $\frac{T_{slowest}}{T_n^{screen}}$ is the speed of client n relative to the slowest client, as determined in the screening round. Since a faster client can complete more rounds in a given amount of time than a slower client, E_n is proportionally related to this speed. Meanwhile, η_n is inversely related to the speed, to allow slower clients to make larger optimization steps, thereby potentially compensating for the fewer training epochs they are able to complete in a given time.

Next, $\frac{D_{\max}}{D_n}$ is the inverse of the ratio of the size of the training set at client n relative to the largest training set among the clients. Clients with large $\frac{D_{\max}}{D_n}$ have less data, therefore, they can complete more training epochs in a given amount of time. Hence, E_n is proportionally related to $\frac{D_{\max}}{D_n}$ in (14). Similarly, clients with less data likely have a less complex error surface than clients with more data, which can be navigated with a higher learning rate. Hence, η_n is also proportionally related to $\frac{D_{\max}}{D_n}$ in (15).

Finally, when discussing packet loss $P_{L,n}$, we should consider two cases: when *loss correction is enabled* and when *loss correction is disabled*. If loss correction is enabled, the SplitFed network can handle missing data using techniques like packet re-transmission or by using packet loss correction methods (tensor completion [53], [54], forward error correction (FEC) [55], etc.). In all these cases, for clients experiencing high $P_{L,n}$, there is a higher latency and/or computational overhead associated with missing data recovery. Hence, fewer epochs can be completed by such clients in a given amount of time. When loss correction is disabled, there is no extra latency or overhead, but there is higher uncertainty in the model updates due to missing data; hence, again, it is prudent to run fewer epochs for clients with large $P_{L,n}$. In summary, E_n should be proportional to $(1 - P_{L,n})$, as shown in (14). For the same reasons, η_n should also be proportional to $(1 - P_{L,n})$, as shown in (15).

To obtain the parameters α , β and γ in (14), we used a least squares estimator from the `np.linalg.lstsq` package [56] under the constraint that $8 \leq E_n \leq 12$. Specifically, we chose the minimum norm (α, β, γ) vector that provided the least squares fit across all clients and all $8 \leq E_n \leq 12$. Thus, one set of (α, β, γ) parameters is obtained, and through (14) they provide E_n for each client.

The parameters δ and σ in (15) are computed through least squares optimization with constraints $1 \times 10^{-5} \leq \eta_n \leq 1 \times 10^{-4}$ and $0 \leq \sigma, \delta \leq 5$. For this, we used the `scipy.optimize` package [57]. This way, one set of (δ, σ) parameters are obtained, and from there, the learning rate η_n is obtained for each client from (15).

IV. EXPERIMENTS

In this section, we provide details of our experimental setup and results. Our experiments include: a performance comparison between SplitFed and AASFL, experimental evidence supporting AASFL's advantage over client selection, statistical significance tests, and comparisons with SOTA approaches.

A. EXPERIMENTAL SETUP

Our proposed training scheme is based on the SplitFed architecture (see Fig. 1), utilizing a Split U-Net [2] model for each client (see Fig. 3). Previous research [49] found that deep splits are more robust to packet loss than shallow splits. Consequently, our experiments in this paper exclusively focus on enhancing the robustness of the shallow split configuration within the SplitFed network.

Our experiments are based on two medical imaging datasets: the Blastocyst dataset [58] and the Human Against Machine with 10000 training images (HAM10K) [59] dataset. The Blastocyst dataset comprises 781 blastocyst samples, while the HAM10K dataset comprises 10,015 dermatoscopic samples. Blastocyst image segmentation is a multi-class task where each sample is segmented into five classes [58], [60]: the trophoctoderm (TE), the zona pellucida (ZP), the blastocoel, the inner cell mass (ICM), and the background component. Dermatoscopic image segmentation is a binary segmentation task where each sample is segmented into two classes [59]: the skin lesion and the background component. First, we horizontally partitioned³ both datasets across five clients, forming a collaborative network with five clients. Second, for a realistic evaluation, we horizontally partitioned only the Blastocyst dataset across seven clients, forming a collaborative network with seven clients. The primary purpose of partitioning the dataset into different numbers of clients was to evaluate the scalability and performance of the AASFL training scheme under varying configurations. By comparing the results from a five-client network to those from a seven-client network, we aimed to assess how the number of participants influences factors such as data distribution efficiency, communication overhead, and overall system performance. We used three different data distributions: non-uniform (NU), uniform (U), and highly non-uniform (HNU). In a NU distribution, clients' datasets are of different sizes, leading to a data distribution imbalance. In a U distribution, every client has the same data size. In a HNU distribution, clients' data show an extreme imbalance, where some clients may have a disproportionately large amount of data while others have very limited data. All three data distributions are realistic in healthcare. Table 1 describes the data distributions used in our experiments. We allocated 70 samples from the Blastocyst dataset and 1010 samples from the HAM10K dataset as the global test set.

In each segmentation task, clients randomly used 85% of the data for training and 15% for validation. Several augmentation processes including horizontal and vertical flipping, rotating, RGB shifting, normalizing, random brightness contrasting were used to improve the quality of the training. *Soft Dice Loss* [61] was used as the loss function with Adam optimizer [62]. Standard SplitFed models were trained with a learning rate of 1×10^{-4} , which was determined based on a balance between convergence speed and training stability, as identified in our prior research [49], [63]. Standard SplitFed models were trained for 10 com rounds, while each client trained their local models for 12 local epochs [48], [49]. For the simulation purpose, packet loss was assumed to be independent and identically distributed (IID) with probability $P_L \in \{0.0, 0.1, 0.3, 0.5, 0.7, 0.9\}$. Each

³Horizontal data partitioning is a random process where data samples are allocated to clients without considering their image data or GTs. Each client holds a portion of the dataset, typically with non-overlapping data samples.

TABLE 1. Client-wise data distribution. U: uniform, NU: non-uniform, HNU: highly non-uniform, C#: the client index, D-Dis: Data Distribution, T: number of training samples, V: number of validation samples.

Dataset	D-Dis	C#	U	NU	HNU
Blastocyst dataset	Collaborative network with 5 clients				
		C1	T:122, V:21	T:204, V:36	T:318, V:55
		C2	T:122, V:20	T:102, V:18	T:102, V:18
		C3	T:122, V:20	T:73, V:12	T:102, V:18
		C4	T:122, V:20	T:152, V:27	T:152, V:27
		C5	T:122, V:20	T:74, V:13	T:3, V:1
	Global test set - 70				
	Collaborative network with 7 clients				
		C1	T:85, V:17	T:170, V:34	T:370, V:73
		C2	T:85, V:17	T:20, V:4	T:5, V:1
		C3	T:85, V:17	T:43, V:9	T:43, V:9
		C4	T:85, V:17	T:100, V:18	T:20, V:4
		C5	T:85, V:17	T:60, V:10	T:5, V:1
		C6	T:85, V:17	T:117, V:25	T:67, V:12
		C7	T:85, V:17	T:85, V:16	T:85, V:16
	Global test set - 70				
HAM10K dataset	Collaborative network with 5 clients				
		C1	T:1531, V:270	T:1603, V:282	T:5377, V:948
		C2	T:1531, V:270	T:1801, V:318	T:205, V:36
		C3	T:1531, V:270	T:1453, V:258	T:61, V:10
		C4	T:1531, V:270	T:2005, V:354	T:2005, V:354
		C5	T:1531, V:270	T:793, V:138	T:7, V:2
	Global test set - 1010				

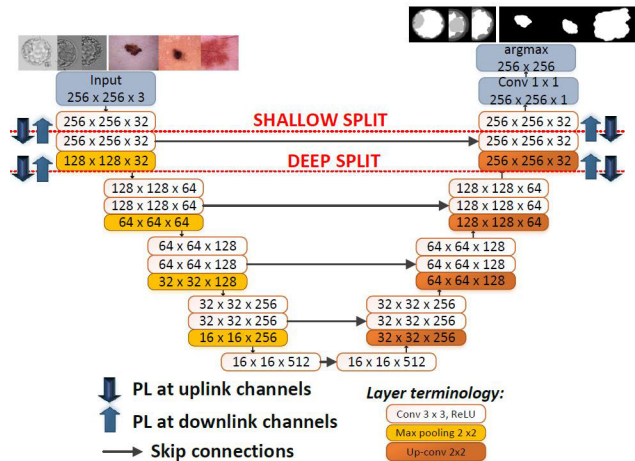


FIGURE 3. Split U-Net model for blastocyst image segmentation and skin lesion segmentation. U-Net model is split into three parts with the help of two split points. Two methods of splitting are decided: the shallow split and the deep split. Features and gradients passing through the split points are subjected to packet loss. PL refers to packet loss.

experiment was run for 10 runs, and the average MJI was recorded.

B. EXPERIMENTAL RESULTS

1) SPLITFED VERSUS AASFL

We compare the average MJIs resulting from the AASFL and SplitFed separately for the two datasets and for the three different data distributions of NU, U, and HNU. Fig. 4 shows the SplitFed vs. AASFL comparison of the average MJI for the Blastocyst dataset, with the collaborative network of five clients, while Fig. 8 shows the corresponding results for the HAM10K dataset. In both of these figures, six curves can

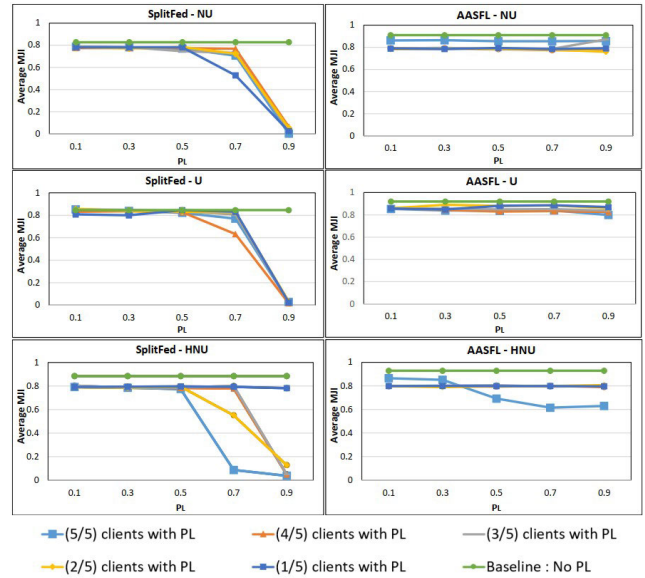


FIGURE 4. Performance comparison of SplitFed vs. AASFL based on average MJI: Blastocyst dataset, collaborative network with 5 clients. Top-to-bottom rows refer to NU, U, and HNU data distributions. PL refers to packet loss.

be seen, which illustrate the average MJI with a varying number of clients experiencing packet loss. Top-to-bottom plots indicate the NU, U, and HNU data distributions. The left-side plots are for the SplitFed, and the right-side plots are for the AASFL. In both Fig. 4 and Fig. 8, it can be seen that SplitFed has failed to maintain a consistent MJI when the packet loss increases. However, AASFL has been able to provide a consistent MJI across different levels of packet loss.

We expanded Fig. 4 into Fig. 5, Fig. 6, and Fig. 7 to get a more detailed representation. We also expanded the x-axis to show the number of clients with packet loss. The four curves in all of these three figures illustrate SplitFed, AASFL, baseline SplitFed and baseline AASFL. In all three figures, it is visible that AASFL's average MJI is higher than that of SplitFed in all cases of packet loss. Similarly, we divided Fig. 8 into Fig. 9, Fig. 10, Fig. 11. The same observation could be seen, where AASFL's average MJI is higher than that of SplitFed in every case with packet loss. Similarly, Fig. 12 shows the SplitFed vs. AASFL comparison of the average MJI for the Blastocyst dataset, with the collaborative network of seven clients.

2) AASFL's ADVANTAGE OVER CLIENT SELECTION

In Section II-D, the importance of client selection along with the identified limitations were discussed. One idea we wish to emphasize is that client selection may not always be beneficial in collaborative networks. We conducted some experiments to verify this idea. SplitFed was run by selecting only the R clients, excluding the participation of the NR clients over the U distribution of the Blastocyst dataset. The resulting MJIs were compared to those resulting from the AASFL. As shown in Fig. 13, the AASFL's MJI was much

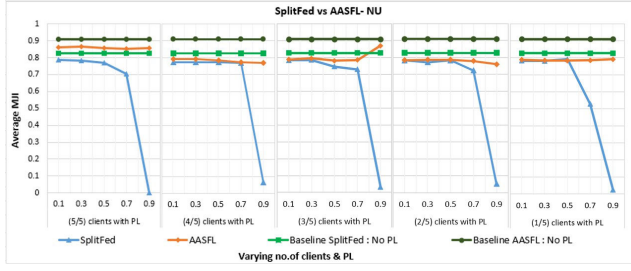


FIGURE 5. Performance comparison of SplitFed vs. AASFL based on average MJJ: non-uniformly distributed Blastocyst dataset, collaborative network with 5 clients. PL refers to packet loss probability.

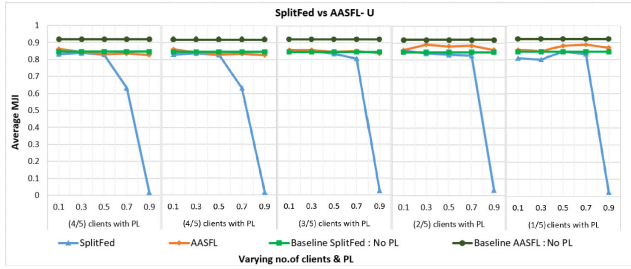


FIGURE 6. Performance comparison of SplitFed vs. AASFL based on average MJJ: uniformly distributed Blastocyst dataset, collaborative network with 5 clients. PL refers to packet loss probability.

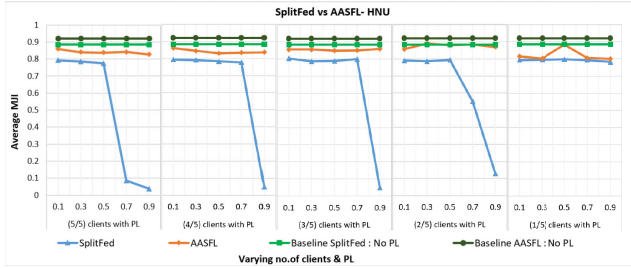


FIGURE 7. Performance comparison of SplitFed vs. AASFL based on average MJJ: highly non-uniformly distributed Blastocyst dataset, collaborative network with 5 clients. PL refers to packet loss probability.

higher than the SplitFed's MJJ when it was trained only with the R clients' participation. Therefore, it can be concluded that AASFL shows comparatively good performance by participating every client in the collaborative training compared to that of SplitFed trained with a client selection mechanism. These experiments were conducted only once; therefore, the y-axis is labeled as 'MJJ', distinguishing it from other figures.

3) STATISTICAL SIGNIFICANT TESTING

As we stated in the Section IV-B1, AASFL results in a higher average MJJ compared to that of SplitFed. We performed a one-tailed t-test [64], to statistically prove this. The one-tailed t-test is the statistical test in which the critical area of a distribution is one-sided so that it is either greater than or less than a certain value, but not both. If the sample being tested falls into the one-sided critical area, the alternative hypothesis will be accepted instead of the null hypothesis. J_{SplitFed} and

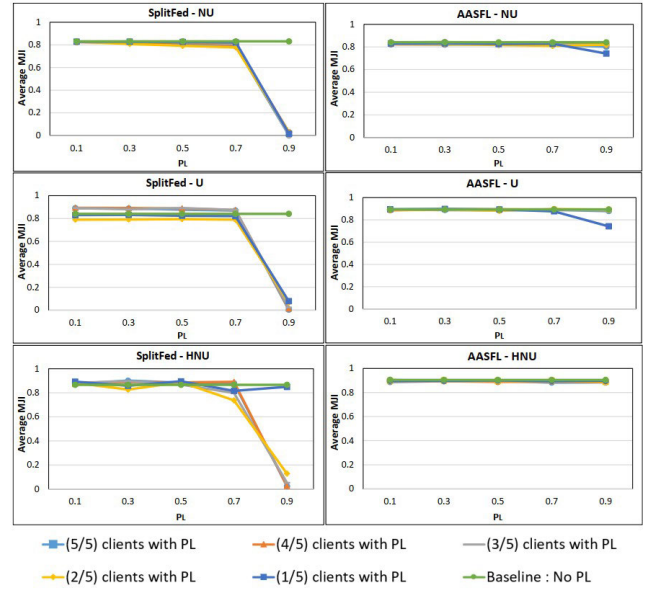


FIGURE 8. Performance comparison of SplitFed vs. AASFL: HAM10K dataset for the collaborative network with five clients. PL refers to packet loss probability.

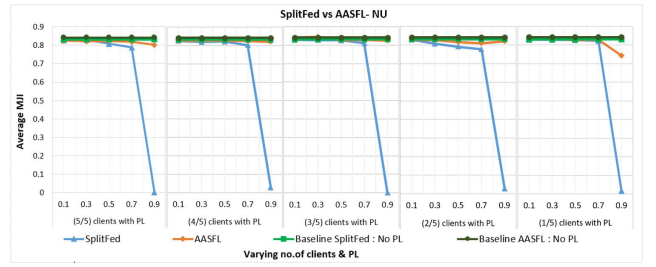


FIGURE 9. Performance comparison of SplitFed vs. AASFL based on average MJJ: non-uniformly distributed HAM10K dataset, collaborative network with 5 clients. PL refers to packet loss probability.

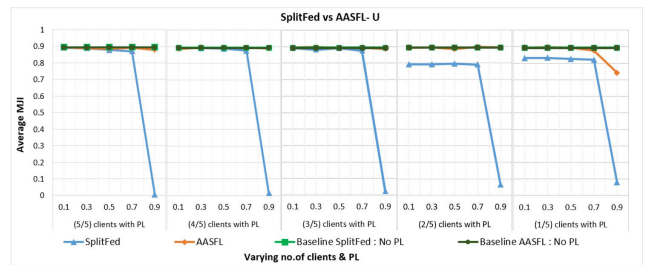


FIGURE 10. Performance comparison of SplitFed vs. AASFL based on average MJJ: uniformly distributed HAM10K dataset, collaborative network with 5 clients. PL refers to packet loss probability.

J_{AASFL} are the average MJJs resulted from SplitFed and AASFL. Our null hypothesis (H_0) is that the two average MJJs are similar, while the alternative hypothesis (H_1) is that AASFL results in a higher MJJ compared to that of SplitFed. The one-tailed t-test is:

$$\begin{aligned} H_0 : J_{\text{AASFL}} &\leq J_{\text{SplitFed}} \\ H_1 : J_{\text{AASFL}} &> J_{\text{SplitFed}} \end{aligned} \quad (18)$$

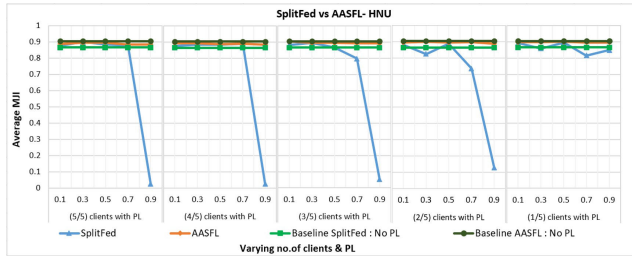


FIGURE 11. Performance comparison of SplitFed vs. AASFL based on average MJJ: highly non-uniformly distributed HAM10K dataset, collaborative network with 5 clients. PL refers to packet loss probability.

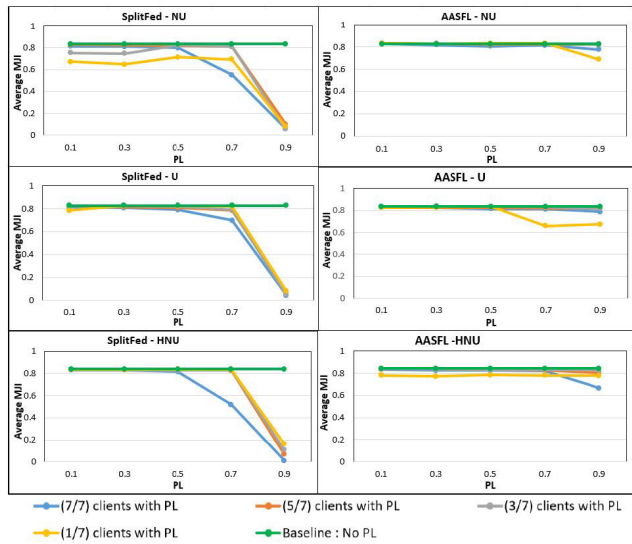


FIGURE 12. Performance comparison of SplitFed vs. AASFL based on average MJJ: Blastocyst dataset for the collaborative network with 7 clients. PL refers to packet loss.

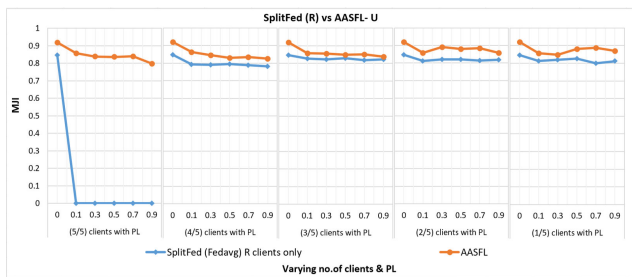


FIGURE 13. Comparing the performance of SplitFed utilizing only R clients vs. AASFL based on MJJ over the uniform distribution of the Blastocyst dataset for the collaborative network with five clients. PL refers to packet loss.

With each dataset, when the collaborative network has five clients, for the three data distribution methods, $\binom{3}{2} = 3$ pairwise comparisons can be made for the six values of P_L and five numbers of clients experiencing packet loss; hence, $3 \times 6 \times 5 = 90$ comparisons. We performed a t-test for each of the 90 cases. Table 2 shows the summary of the pairwise t-tests between SplitFed and AASFL over the Blastocyst dataset, while Table 3 shows the corresponding

TABLE 2. Summary of the pairwise t-tests between SplitFed and AASFL over the Blastocyst dataset with the collaborative network of five clients. U: uniform, NU: non-uniform, HNU: highly non-uniform, D-Dis: Data Distribution, #C: Number of clients with packet loss, Values less than 0.05 are highlighted in green.

		One-tail p-value					
D-Dis	#C	rounded off to 2 nd decimal place					
P_L		0.0	0.1	0.3	0.5	0.7	0.9
NU	5	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	0.00	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.00	0.00	0.00	0.00
	2	0.00	0.19	0.00	0.00	0.00	0.00
	1	0.00	0.00	0.00	0.00	0.00	0.00
U	5	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	0.00	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.00	0.00	0.00	0.00
	2	0.00	0.39	0.00	0.00	0.00	0.00
	1	0.00	0.00	0.00	0.00	0.00	0.00
HNU	5	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	0.00	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.00	0.00	0.00	0.00
	2	0.00	0.00	0.00	0.00	0.00	0.00
	1	0.00	0.00	0.00	0.00	0.00	0.00

values over the HAM10K dataset. In both of these tables, the cells with one-tailed p value [65] < 0.05 are highlighted in green. In Table 2, 97.78%(88/90) of the t-tests we have $p < 0.05$, while in Table 3, 93.34%(84/90) of the t-tests we have $p < 0.05$. Therefore, in both segmentation tasks, one can reject the null-hypothesis H_0 and conclude that AASFL produces a higher MJJ than SplitFed at the significance level of 0.05. It was also identified that in instances where $p > 0.05$ are those from the cases where SplitFed's MJJ is closer or a bit higher than that of AASFL's MJJ. These cases can be interpreted as the packet loss acting as a dropout [49] in SplitFed.

Similarly, when the collaborative network has seven clients, evaluating only for four scenarios of clients experiencing packet loss; we have $3 \times 6 \times 4 = 72$ comparisons. Table 4 shows the summary of the pairwise t-tests between SplitFed and AASFL over the Blastocyst dataset. Of all the t-tests we have $p < 0.05$, therefore, one can reject the null-hypothesis H_0 and conclude that AASFL produces a higher MJJ than SplitFed at the significance level of 0.05.

4) COMPARISON OVER SOTA APPROACHES

We compare our proposed AASFL training scheme with five well-known SOTA parameter aggregation approaches: naïve averaging (naïve avg) [5], federated averaging (fedAvg) [5], auto-FedAvg [66], fed-NCL_V2 [67], and fed-NCL_V4 [67]. In naïve avg, parameter aggregation is based on the number of clients. In fedAvg, parameter aggregation is based on the clients' data distribution. In auto-FedAvg, parameter aggregation is based on clients' data distribution and current training progress. The fed-NCL_V2 uses network-wise parameter aggregation, while fed-NCL_V2 uses layer-wise parameter aggregation. Parameter aggregation of both

TABLE 3. Summary of the pairwise t-tests between SplitFed and AASFL over the HAM10K dataset with the collaborative network of five clients. U: uniform, NU: non-uniform, HNU: highly non-uniform, D-Dis: Data Distribution, #C: Number of clients with packet loss, Values less than 0.05 are highlighted in green.

D-Dis	#C	One-tail p-value					
		rounded off to 2 nd decimal place					
P_L		0.0	0.1	0.3	0.5	0.7	0.9
NU	5	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	0.00	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.00	0.00	0.00	0.00
	2	0.00	0.00	0.00	0.00	0.00	0.00
	1	0.00	0.00	0.00	0.00	0.00	0.00
U	5	0.00	0.14	0.09	0.00	0.00	0.00
	4	0.00	0.00	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.06	0.00	0.00	0.00
	2	0.00	0.00	0.00	0.00	0.00	0.00
	1	0.00	0.00	0.00	0.00	0.00	0.00
HNU	5	0.00	0.11	0.00	0.00	0.33	0.00
	4	0.00	0.00	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.08	0.00	0.00	0.00
	2	0.00	0.00	0.00	0.00	0.00	0.00
	1	0.00	0.00	0.00	0.00	0.00	0.00

TABLE 4. Summary of the pairwise t-tests between SplitFed and AASFL on Blastocyst dataset with the collaborative network of 7 clients. U: uniform, NU: non-uniform, HNU: highly non-uniform, D-Dis: Data Distribution, #C: Number of clients with packet loss, Values less than 0.05 are highlighted in green.

D-Dis	#C	One-tail p-value					
		rounded off to 2 nd decimal place					
P_L		0.0	0.1	0.3	0.5	0.7	0.9
NU	7	0.00	0.00	0.00	0.00	0.00	0.00
	5	0.00	0.00	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.00	0.00	0.00	0.00
	1	0.00	0.00	0.00	0.00	0.00	0.00
U	7	0.00	0.00	0.00	0.00	0.00	0.00
	5	0.00	0.00	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.00	0.00	0.00	0.00
	1	0.00	0.00	0.00	0.00	0.00	0.00
HNU	7	0.00	0.00	0.00	0.00	0.00	0.00
	5	0.00	0.00	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.00	0.00	0.00	0.00
	1	0.00	0.00	0.00	0.00	0.00	0.00

fed-NCL versions is based on the clients' data distribution and local model divergence from the aggregated global model, while fed-NCL_V2 additionally considers the training loss of local training data on the aggregated global model. The comparison of these five aggregation methods with AASFL for the NU data distribution of the Blastocyst dataset is illustrated in Fig. 14. We observe that AASFL is the best-performing model compared to that of the other five parameter aggregation approaches, even at high packet loss rates. This is due to its adaptive and asynchronous design, which dynamically adjusts clients' training based on communication quality and client data heterogeneity. Unlike other methods, AASFL mitigates synchronization bottlenecks and ensures robust performance even under high

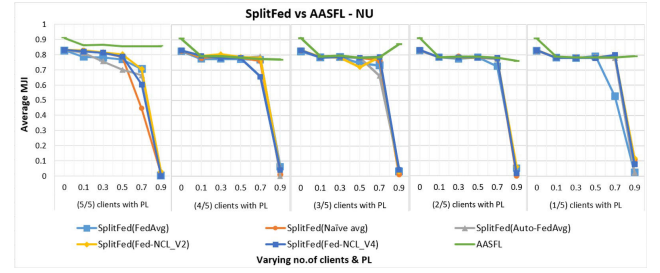


FIGURE 14. Performance comparison of SplitFed on 5 SOTA parameter aggregation approaches vs. AASFL based on average MJJ over the non-uniform distribution of the Blastocyst dataset for the collaborative network with five clients. PL refers to packet loss probability.

packet loss, achieving superior convergence and reliability in the SplitFed setting.

V. CONCLUSION AND FUTURE DIRECTIONS

This paper presented a novel AASFL training scheme for SplitFed learning. The asynchronicity of AASFL benefits clients with clean communication links, while its adaptability ensures clients' adaptive local training. We compared the performance of the proposed AASFL training scheme with standard SplitFed in various scenarios with packet loss. While standard SplitFed remained robust only up to around 50% of packet loss, AASFL showed complete robustness over any packet loss rate. AASFL significantly outperformed five traditional FL parameter aggregation approaches, including standard FedAvg. In addition, we experimentally brought up the idea of preserving each and every client's local training, showing some drawbacks of client selection. Our AASFL training scheme outperformed splitFed trained with client selection. We validated our experiments statistically and theoretically. In conclusion, this unique combination of asynchronicity and adaptability sets AASFL apart as an effective training scheme for SplitFed learning, making it suitable for real-world healthcare applications. As future work, we aim to develop a better understanding of decaying learning rates in the SplitFed network. Concurrently, we intend to provide a theoretical comparison between standard FL and AASFL with the achieved theoretical bounds. Furthermore, we also intend to employ the achieved bounds to predict the hyperparameters of AASFL successfully.

APPENDIX-CONVERGENCE ANALYSIS

In this section, we detail a theoretic explanation of the convergence analysis of the proposed AASFL training scheme. The main goal of proving convergence is to theoretically analyze the feasibility of our approach by showing it achieves a constant convergence bound. Our bounds estimate the performance difference between the current model and the optimal model, in terms of *model weights*. The bounds are expressed in terms of the expected loss of the global model.

A. ASSUMPTIONS AND NOTATION

Following similar analysis of FL [21], [22], [25], [68], [69], [70], [71] and SplitFed [71], [72], we use the same notation

and assumptions. As in Section III-A2, $F(w)$ is the global loss function of the global model, and $f_n(w)$ is the local loss of the local model at client n . To simplify the notation, we will use $\nabla f_n(w) := \mathbb{E}[\nabla_w f_n(w)]$ and $\nabla F(w) := \mathbb{E}[\nabla_w F(w)]$; that is, we will drop the expectation and subscripts from the expected gradients. Also, $\nabla f_n(w)^M$ stands for the gradient after M T -com rounds.

The assumptions are as follows:

- 1) L -smoothness For each client n , $f_n(y) \leq f_n(x) + \langle \nabla f(x), y-x \rangle + \frac{L}{2} \|y-x\|^2, \forall x, y$.
- 2) μ -strong convexity For each client n , $f_n(y) \geq f_n(x) + \langle \nabla f(x), y-x \rangle + \frac{\mu}{2} \|y-x\|^2, \forall x, y$.
- 3) Bounded stochastic gradient square norm For each client n , $\mathbb{E} \|\nabla f_n(w)\|^2 \leq r^2, \forall w$.
- 4) Existence of a global optimum There exists at least one solution, denoted as w^* to achieve the global minimum of $F(w)$.

B. PERFORMANCE BOUNDS

Analyzing the performance bound of AASFL is more challenging than that of conventional FL due to its tri-paced model updates. That is, each client updates the two client side model parts (w_{FE} and w_{BE}) and the server side model (w_S) in a SplitFed manner. Following [72], we address this challenge by decomposing the analysis into three parts: the client side frontend (w_{FE}), the client side backend (w_{BE}), and the server side (w_S) model updates. We describe the decomposition procedure below.

Let $w^* := [w_{T,FE}^*, w_{T,S}^*, w_{T,BE}^*]$ denote the optimal global model that minimizes the expected global loss function $F(w)$. We show that the expected loss of the AASFL training scheme's global model and the optimal global model is upper-bounded by three terms. These terms are related to the client side frontend, server side, and client side backend models, respectively. We formalize this result in Lemma 1.

Lemma 1 (Decomposition): Let w^M be the global model after M T -com rounds. Then we have:

$$\mathbb{E}[F(w^M)] - F(w^*) \leq \frac{L}{2} \left(\mathbb{E} \|w_{T,FE}^M - w_{T,FE}^*\|^2 + \mathbb{E} \|w_{T,S}^M - w_{T,S}^*\|^2 + \mathbb{E} \|w_{T,BE}^M - w_{T,BE}^*\|^2 \right), \quad (19)$$

where L is the L -smoothness constant.

Proof: Since $F(w)$ is a linear combination of clients' loss functions (3), under Assumption 1, it is easy to prove that $F(w)$ is L -smooth. Thus, we have:

$$\begin{aligned} \mathbb{E}[F(w^M)] - F(w^*) &\leq \mathbb{E}[\langle \nabla F(w^*), w^M - w^* \rangle] \\ &\quad + \frac{L}{2} \mathbb{E}[\|w^M - w^*\|^2] \\ &= \frac{L}{2} \mathbb{E}[\|w^M - w^*\|^2]. \end{aligned} \quad (20)$$

Equation 20 is from assumption 4, which we can consider that the difference between w^M and w^* becomes smaller, leading to $w^M - w^*$ approaching zero with the T -com rounds.

Since $w^M := [w_{T,FE}^M; w_{T,S}^M; w_{T,BE}^M]$ and $w^* := [w_{T,FE}^*; w_{T,S}^*; w_{T,BE}^*]$, we have:

$$\mathbb{E}[\|w^M - w^*\|^2] = \mathbb{E} \left[\left\| \begin{bmatrix} w_{T,FE}^M \\ w_{T,S}^M \\ w_{T,BE}^M \end{bmatrix} - \begin{bmatrix} w_{T,FE}^* \\ w_{T,S}^* \\ w_{T,BE}^* \end{bmatrix} \right\|^2 \right]. \quad (21)$$

Rewriting,

$$\mathbb{E}[\|w^M - w^*\|^2] = \mathbb{E} \left[\left\| \begin{bmatrix} w_{T,FE}^M - w_{T,FE}^* \\ w_{T,S}^M - w_{T,S}^* \\ w_{T,BE}^M - w_{T,BE}^* \end{bmatrix} \right\|^2 \right]. \quad (22)$$

Expressing the squared Euclidean norm component wise:

$$\begin{aligned} \mathbb{E}[\|w^M - w^*\|^2] &= \mathbb{E}[\|w_{T,FE}^M - w_{T,FE}^*\|^2] \\ &\quad + \mathbb{E}[\|w_{T,S}^M - w_{T,S}^*\|^2] \\ &\quad + \mathbb{E}[\|w_{T,BE}^M - w_{T,BE}^*\|^2]. \end{aligned} \quad (23)$$

Using this in (20):

$$\begin{aligned} \mathbb{E}[F(w^M)] - F(w^*) &\leq \frac{L}{2} \left(\mathbb{E}[\|w_{T,FE}^M - w_{T,FE}^*\|^2] \right. \\ &\quad + \mathbb{E}[\|w_{T,S}^M - w_{T,S}^*\|^2] \\ &\quad \left. + \mathbb{E}[\|w_{T,BE}^M - w_{T,BE}^*\|^2] \right). \end{aligned} \quad (24)$$

This completes the proof of Lemma 1. ■

To limit the loss of the global model, Lemma 1 says that we only need to limit the expected gap between the models from the AASFL and the optimal model at the client side frontend, the server side, and the client side backend.

Next, we bound the expected model gap at the BE, S, and FE during their local training, i.e., $\mathbb{E}[\|w_{BE}^{E_n} - w_{BE}^*\|^2]$, $\mathbb{E}[\|w_S^{E_n} - w_S^*\|^2]$, and $\mathbb{E}[\|w_{FE}^{E_n} - w_{FE}^*\|^2]$, in Theorem 1. E_n and η_n are the adaptive local epochs and the adaptive learning rates that we find in equations 14 and 15, respectively.

Theorem 1: Under assumptions 1-4, the upper bounds on the expected deviation of BE, S, and FE models for client n from the optimal models over E_n local epochs with η_n learning rate are given by:

$$\begin{aligned} \mathbb{E}[\|w_{BE}^{E_n} - w_{BE}^*\|^2] &\leq (1 - \mu\eta_n)^{E_n} [\|w_{BE}^0 - w_{BE}^*\|^2] \\ &\quad + \frac{2\eta_n^2 r^2 E_n}{N^2}, \end{aligned} \quad (25)$$

$$\begin{aligned} \mathbb{E}[\|w_S^{E_n} - w_S^*\|^2] &\leq (1 - \mu\eta_n)^{E_n} [\|w_S^0 - w_S^*\|^2] \\ &\quad + \frac{2\eta_n^2 r^2 E_n}{N^2}, \end{aligned} \quad (26)$$

$$\begin{aligned} \mathbb{E}[\|w_{FE}^{E_n} - w_{FE}^*\|^2] &\leq (1 - \mu\eta_n)^{E_n} [\|w_{FE}^0 - w_{FE}^*\|^2] \\ &\quad + \frac{2\eta_n^2 r^2 E_n}{N^2}, \end{aligned} \quad (27)$$

where w_{BE}^0 , w_S^0 , and w_{FE}^0 are the initialized client side backend, server side and client side front end models, respectively.

Proof: First, we bound the expected model gap at the client side backend, i.e., $\mathbb{E}[\|w_{BE}^M - w_{BE}^*\|^2]$. e represents an epoch of a particular client n , where $e \in E_n$. η_n is the learning rate of a particular client n .

$$\begin{aligned} & \mathbb{E}[\|w_{BE}^e - w_{BE}^*\|^2] \\ &= \mathbb{E}\left[\left\|w_{BE}^{e-1} - \eta_n \frac{1}{N} \sum_{n=1}^N \nabla_{w_{BE}} f_n(w^{e-1}) - w_{BE}^*\right\|^2\right] \end{aligned} \quad (28)$$

Expanding:

$$\begin{aligned} \mathbb{E}[\|w_{BE}^e - w_{BE}^*\|^2] &= \mathbb{E}[\|w_{BE}^{e-1} - w_{BE}^*\|^2] \\ &\quad - 2\eta_n \mathbb{E}[\langle \nabla_{w_{BE}} F(w^{e-1}), w_{BE}^{e-1} - w_{BE}^* \rangle] \\ &\quad + \eta_n^2 \mathbb{E}\left[\left\|\frac{1}{N} \sum_{n=1}^N \nabla_{w_{BE}} f_n(w^{e-1})\right\|^2\right] \end{aligned} \quad (29)$$

$$\begin{aligned} &\leq (1 - \mu\eta_n) \mathbb{E}[\|w_{BE}^{e-1} - w_{BE}^*\|^2] - 2\eta_n \mathbb{E}[F(w^{e-1}) \\ &\quad - F(w_S^{e-1}, w_{BE}^*)] + \eta_n^2 \mathbb{E}\left[\left\|\frac{1}{N} \sum_{n=1}^N \nabla_{w_{BE}} f_n(w^{e-1})\right\|^2\right] \end{aligned} \quad (30)$$

$$\begin{aligned} &\leq (1 - \mu\eta_n) \mathbb{E}[\|w_{BE}^{e-1} - w_{BE}^*\|^2] - 2\eta_n \mathbb{E}[F(w^{e-1}) \\ &\quad - F(w_S^{e-1}, w_{BE}^*)] + \eta_n^2 \mathbb{E}C_1, \end{aligned} \quad (31)$$

where $F(w^{e-1}) = F(w_S^{e-1}, w_{BE}^{e-1})$ and $C_1 = \mathbb{E}[\left\|\frac{1}{N} \sum_{n=1}^N \nabla_{w_{BE}} f_n(w^{e-1})\right\|^2]$.

Next, we bound C_1 as follows:

$$\begin{aligned} C_1 &= \mathbb{E}\left[\left\|\frac{1}{N} \sum_{n=1}^N \nabla_{w_{BE}} f_n(w^{e-1}) - \nabla_{w_{BE}} f_n(w^*)\right\|^2\right] \\ &\quad + \mathbb{E}[\|\nabla_{w_{BE}} f_n(w^*)\|^2] \end{aligned} \quad (32)$$

$$\begin{aligned} &\leq 2 \mathbb{E}\left[\left\|\frac{1}{N} \sum_{n=1}^N [\nabla_{w_{BE}} f_n(w^{e-1}) - \nabla_{w_{BE}} f_n(w^*)]\right\|^2\right] \\ &\quad + 2 \mathbb{E}\left[\left\|\frac{1}{N} \sum_{n=1}^N \nabla_{w_{BE}} f_n(w^*)\right\|^2\right] \end{aligned} \quad (33)$$

$$\begin{aligned} &\leq \frac{2}{N} \sum_{n=1}^N \mathbb{E}\left[\left\|\sum_{n=1}^N [\nabla_{w_{BE}} f_n(w^{e-1}) - \nabla_{w_{BE}} f_n(w^*)]\right\|^2\right] \\ &\quad + \frac{2}{N} \mathbb{E}\left[\left\|\sum_{n=1}^N \nabla_{w_{BE}} f_n(w^*)\right\|^2\right] \end{aligned} \quad (34)$$

$$\begin{aligned} &\leq \frac{2}{N} \sum_{n=1}^N \mathbb{E}\left[\left\|\nabla_{w_{BE}} f_n(w^{e-1}) - \nabla_{w_{BE}} f_n(w^*)\right\|^2\right] \\ &\quad + \frac{2r^2}{N^2} \\ &\leq \frac{4L}{N} \sum_{n=1}^N \mathbb{E}[f_n(w^{e-1}) - f_n(w^*)] \end{aligned} \quad (35)$$

$$- \langle \nabla_{w_{BE}} f_n(w^*), w^{e-1} - w^* \rangle] + \frac{2r^2}{N^2} \quad (36)$$

$$= 4L \mathbb{E}[F(w_S^{e-1}, w_{BE}^{e-1}) - F(w_S^{e-1}, w_{BE}^*)] + \frac{2r^2}{N^2}, \quad (37)$$

where

Equation (33) is from the inequality: $(x - y + y)^2 \leq 2(x - y)^2 + 2y^2$.

Equation (34) is from the inequality: $\|\sum_{n=1}^N a_n\|^2 \leq N \sum_{n=1}^N \|a_n\|^2$.

Equation (35) is from Assumption 3, and the fact that:

$$\mathbb{E}\left[\left\|\sum_{n=1}^N \nabla_{w_{BE}} f_n(w^*)\right\|^2\right] \leq \mathbb{E}\left[\sum_{n=1}^N \|\nabla_{w_{BE}} f_n(w^*)\|^2\right].$$

Equation (36) is from Lemma 1 and Assumption 4.

Plugging (37) to (31),

$$\begin{aligned} \mathbb{E}[\|w_{BE}^e - w_{BE}^*\|^2] &\leq (1 - \mu\eta_n) \mathbb{E}[\|w_{BE}^{e-1} - w_{BE}^*\|^2] \\ &\quad - 2\eta_n \mathbb{E}[F(w^{e-1}) - F(w_S^{e-1}, w_{BE}^*)] \\ &\quad + 4L\eta_n^2 \{ \mathbb{E}[F(w_S^{e-1}, w_{BE}^{e-1}) \\ &\quad - F(w_S^{e-1}, w_{BE}^*)] + \frac{2r^2}{N^2} \} \end{aligned} \quad (38)$$

$$\begin{aligned} &\leq (1 - \mu\eta_n) \mathbb{E}[\|w_{BE}^{e-1} - w_{BE}^*\|^2] \\ &\quad - 2\eta_n (1 - 2L\eta_n) \mathbb{E}[F(w_S^{e-1}, w_{BE}^{e-1}) - F(w_S^{e-1}, w_{BE}^*)] \\ &\quad + \frac{2\eta_n^2 r^2}{N^2} \end{aligned} \quad (39)$$

$$\begin{aligned} &\leq (1 - \mu\eta_n) \mathbb{E}[\|w_{BE}^{e-1} - w_{BE}^*\|^2] - \eta_n \mathbb{E}[F(w_S^{e-1}, w_{BE}^{e-1}) \\ &\quad - F(w_S^{e-1}, w_{BE}^*)] + \frac{2\eta_n^2 r^2}{N^2}. \end{aligned} \quad (40)$$

Let $\Delta^{e+1} := \mathbb{E}[\|w_{BE}^e - w_{BE}^*\|^2]$, then we can rewrite equation (40) as:

$$\begin{aligned} \Delta^{e+1} &\leq (1 - \mu\eta_n) \Delta^e - \eta_n \mathbb{E}[F(w_S^{e-1}, w_{BE}^{e-1}) \\ &\quad - F(w_S^{e-1}, w_{BE}^*)] + \frac{2\eta_n^2 r^2}{N^2}. \end{aligned} \quad (41)$$

Since $F(w_S^{e-1}, w_{BE}^{e-1}) \geq F(w_S^{e-1}, w_{BE}^*)$ and $\eta_n > 0$, we have

$$\Delta^{e+1} \leq (1 - \mu\eta_n) \Delta^e + \frac{2\eta_n^2 r^2}{N^2}. \quad (42)$$

Using $e = E_n$ and $e = E_n - 1$ in (42), we get, respectively,

$$\Delta^{E_n+1} \leq (1 - \mu\eta_n) \Delta^{E_n} + \frac{2\eta_n^2 r^2}{N^2}, \quad (43)$$

$$\Delta^{E_n} \leq (1 - \mu\eta_n) \Delta^{E_n-1} + \frac{2\eta_n^2 r^2}{N^2} \quad (44)$$

Substituting (44) into (43) we obtain:

$$\begin{aligned} \Delta^{E_n+1} &\leq (1 - \mu\eta_n) \left[(1 - \mu\eta_n) \Delta^{E_n-1} + \frac{2\eta_n^2 r^2}{N^2} \right] \\ &\quad + \frac{2\eta_n^2 r^2}{N^2} \end{aligned} \quad (45)$$

Continuing to telescope by substituting Δ^{e+1} from (42) with $e = E_n - 2, E_n - 3, \dots, 1$, we get

$$\Delta^{E_n+1} \leq (1 - \mu\eta_n)^{E_n} \Delta^1 + \frac{2\eta_n^2 r^2}{N^2} \sum_{t=1}^{E_n} (1 - \mu\eta_n)^{t-1}$$

$$\begin{aligned}
&= (1 - \mu\eta_n)^{E_n} \Delta^1 + \frac{2\eta_n^2 r^2}{N^2} \cdot \frac{E_n \mu \eta_n}{1 - (1 - \mu\eta_n)} \\
&= (1 - \mu\eta_n)^{E_n} \Delta^1 + \frac{2\eta_n^2 r^2}{N^2} E_n. \quad (46)
\end{aligned}$$

Finally, from the fact that $\Delta^{e+1} = \mathbb{E}[\|w_{BE}^e - w_{BE}^*\|^2]$, we get

$$\begin{aligned}
\mathbb{E}[\|w_{BE}^{E_n} - w_{BE}^*\|^2] &\leq (1 - \mu\eta_n)^{E_n} [\|w_{BE}^0 - w_{BE}^*\|^2] \\
&\quad + \frac{2\eta_n^2 r^2 E_n}{N^2}. \quad (47)
\end{aligned}$$

Similarly,

$$\begin{aligned}
\mathbb{E}[\|w_S^{E_n} - w_S^*\|^2] &\leq (1 - \mu\eta_n)^{E_n} [\|w_S^0 - w_S^*\|^2] \\
&\quad + \frac{2\eta_n^2 r^2 E_n}{N^2}. \quad (48)
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}[\|w_{FE}^{E_n} - w_{FE}^*\|^2] &\leq (1 - \mu\eta_n)^{E_n} [\|w_{FE}^0 - w_{FE}^*\|^2] \\
&\quad + \frac{2\eta_n^2 r^2 E_n}{N^2}. \quad (49)
\end{aligned}$$

Theorem 1 is proved. ■

Next, we bound the expected model gap at the BE, S, and FE during R clients' local training, i.e., $\mathbb{E}[\|w_{R,BE}^K - w_{BE}^*\|^2]$, $\mathbb{E}[\|w_{R,S}^K - w_S^*\|^2]$, and $\mathbb{E}[\|w_{R,FE}^K - w_{FE}^*\|^2]$, in Theorem 2. N_R is the number of R clients. As previously mentioned, K is the number of R -com rounds.

Theorem 2: Under Assumptions 1-4, the upper bounds on the expected deviation of BE, S, and FE models for client n from the optimal models during R clients' local training and their parameter aggregation are given by:

$$\begin{aligned}
&\mathbb{E}[\|w_{R,BE}^K - w_{BE}^*\|^2] \\
&\leq \sum_{n=1}^{N_R} p_{R,n} (1 - \mu\eta_n)^{E_n K} [\|w_{BE}^0 - w_{BE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n K}{N^2}, \quad (50)
\end{aligned}$$

$$\begin{aligned}
&\mathbb{E}[\|w_{R,S}^K - w_S^*\|^2] \\
&\leq \sum_{n=1}^{N_R} p_{R,n} (1 - \mu\eta_n)^{E_n K} [\|w_S^0 - w_S^*\|^2] + \frac{2\eta_n^2 r^2 E_n K}{N^2}, \quad (51)
\end{aligned}$$

$$\begin{aligned}
&\mathbb{E}[\|w_{R,FE}^K - w_{FE}^*\|^2] \\
&\leq \sum_{n=1}^{N_R} p_{R,n} (1 - \mu\eta_n)^{E_n K} [\|w_{FE}^0 - w_{FE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n K}{N^2}, \quad (52)
\end{aligned}$$

where $p_{R,n}$ is the normalized data ratio of the n^{th} R client, that was defined under equation (2).

Proof: In each R -com round, each R client performs E_n local epochs. Since, R clients collaboratively conduct K rounds of R -com rounds, following Theorem 1, we can

express the expected model gap for R clients at BE, S, and FE as follows:

$$\begin{aligned}
\mathbb{E}[\|w_{BE}^{E_n K} - w_{BE}^*\|^2] &\leq (1 - \mu\eta_n)^{E_n K} [\|w_{BE}^0 - w_{BE}^*\|^2] \\
&\quad + \frac{2\eta_n^2 r^2 E_n K}{N^2}. \quad (53)
\end{aligned}$$

Similarly,

$$\begin{aligned}
\mathbb{E}[\|w_S^{E_n K} - w_S^*\|^2] &\leq (1 - \mu\eta_n)^{E_n K} [\|w_S^0 - w_S^*\|^2] \\
&\quad + \frac{2\eta_n^2 r^2 E_n K}{N^2}. \quad (54)
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}[\|w_{FE}^{E_n K} - w_{FE}^*\|^2] &\leq (1 - \mu\eta_n)^{E_n K} [\|w_{FE}^0 - w_{FE}^*\|^2] \\
&\quad + \frac{2\eta_n^2 r^2 E_n K}{N^2}. \quad (55)
\end{aligned}$$

After each R -com round, R clients' model weights are aggregated based on $p_{R,n}$ of R clients. $w_{BE}^{E_n K}$ is the BE model created by the n^{th} R client after $E_n K$ local epochs. We can express this aggregation as:

$$w_{R,BE}^K = \sum_{n=1}^{N_R} p_{R,n} w_{BE}^{E_n K}. \quad (56)$$

Let $d_{BE}^n = w_{BE}^{E_n K} - w_{BE}^*$ be the distance of the n^{th} R client's weights from the optimal weights at the end of K R -com rounds, and let $d_{BE} = \sum_{n=1}^{N_R} p_{R,n} d_{BE}^n$. Since $p_{R,n}$'s are non-negative and add up to 1, d_{BE} is a convex combination of d_{BE}^n 's, and we can use the well-known inequality about the norm of a convex combination of vectors [73] to conclude that $\|d_{BE}\| \leq \sum_{n=1}^{N_R} p_{R,n} \|d_{BE}^n\|$. Substituting the definitions of d_{BE}^n and d_{BE} , and taking the expectation, we conclude:

$$\mathbb{E}[\|w_{R,BE}^K - w_{BE}^*\|^2] \leq \sum_{n=1}^{N_R} p_{R,n} \mathbb{E}[\|w_{BE}^{E_n K} - w_{BE}^*\|^2]. \quad (57)$$

Substituting the individual bound we got in (53) into (57) we obtain:

$$\begin{aligned}
&\mathbb{E}[\|w_{R,BE}^K - w_{BE}^*\|^2] \\
&\leq \sum_{n=1}^{N_R} p_{R,n} (1 - \mu\eta_n)^{E_n K} [\|w_{BE}^0 - w_{BE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n K}{N^2}. \quad (58)
\end{aligned}$$

Similarly, substituting the individual bound in (54) into (57):

$$\begin{aligned}
\mathbb{E}[\|w_{R,S}^K - w_S^*\|^2] &\leq \sum_{n=1}^{N_R} p_{R,n} (1 - \mu\eta_n)^{E_n K} [\|w_S^0 - w_S^*\|^2] \\
&\quad + \frac{2\eta_n^2 r^2 E_n K}{N^2}. \quad (59)
\end{aligned}$$

and, substituting the individual bound in (55) into (57):

$$\mathbb{E}[\|w_{R,FE}^K - w_{FE}^*\|^2]$$

$$\leq \sum_{n=1}^{N_R} p_{R,n}(1 - \mu\eta_n)^{E_n K} [\|w_{FE}^0 - w_{FE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n K}{N^2}. \quad (60)$$

Theorem 2 is proved. ■

Next, we bound the expected model gap at the BE, S, and FE during N_R clients' local training, i.e., $\mathbb{E}[\|w_{NR, BE}^{E_n} - w_{BE}^*\|^2]$, $\mathbb{E}[\|w_{NR, S}^{E_n} - w_S^*\|^2]$, and $\mathbb{E}[\|w_{NR, FE}^{E_n} - w_{FE}^*\|^2]$, in Theorem 3. N_{NR} is the number of N_R clients.

Theorem 3: Under Assumptions 1-4, the upper bounds on the expected deviation of BE, S, and FE models for client n from the optimal models during N_R clients' local training and their parameter aggregation are given by:

$$\begin{aligned} & \mathbb{E}[\|w_{NR, BE}^{E_n} - w_{BE}^*\|^2] \\ & \leq \sum_{n=1}^{N_{NR}} p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_{BE}^0 - w_{BE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2}, \end{aligned} \quad (61)$$

$$\begin{aligned} & \mathbb{E}[\|w_{NR, S}^{E_n} - w_S^*\|^2] \\ & \leq \sum_{n=1}^{N_{NR}} p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_S^0 - w_S^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2}, \end{aligned} \quad (62)$$

$$\begin{aligned} & \mathbb{E}[\|w_{NR, FE}^{E_n} - w_{FE}^*\|^2] \\ & \leq \sum_{n=1}^{N_{NR}} p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_{FE}^0 - w_{FE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2}, \end{aligned} \quad (63)$$

where $p_{NR,n}$ is the modified data ratio of the n^{th} N_R client, that was defined under equation (17).

Proof: In one T -com round, each N_R client performs E_n local epochs. After all N_R clients finish their local training, their model weights are aggregated based on $p_{NR,n}$ of N_R clients. $w_{BE}^{E_n}$ is the BE model created by the n^{th} N_R client after E_n local epochs. We can express this aggregation as:

$$w_{NR, BE}^{E_n} = \sum_{n=1}^{N_{NR}} p_{NR,n} w_{BE}^{E_n}. \quad (64)$$

Similar in deriving (57) in Theorem 2, using the norm of a combination of vectors, we conclude:

$$\mathbb{E}[\|w_{NR, BE}^{E_n} - w_{BE}^*\|^2] \leq \sum_{n=1}^{N_{NR}} p_{NR,n} \mathbb{E}[\|w_{BE}^{E_n} - w_{BE}^*\|^2]. \quad (65)$$

Telescoping and substituting from Theorem 1, we obtain:

$$\begin{aligned} & \mathbb{E}[\|w_{NR, BE}^{E_n} - w_{BE}^*\|^2] \\ & \leq \sum_{n=1}^{N_{NR}} p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_{BE}^0 - w_{BE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2}. \end{aligned} \quad (66)$$

Similarly,

$$\begin{aligned} & \mathbb{E}[\|w_{NR, S}^{E_n} - w_S^*\|^2] \\ & \leq \sum_{n=1}^{N_{NR}} p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_S^0 - w_S^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2}. \end{aligned} \quad (67)$$

and,

$$\begin{aligned} & \mathbb{E}[\|w_{NR, FE}^{E_n} - w_{FE}^*\|^2] \\ & \leq \sum_{n=1}^{N_{NR}} p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_{FE}^0 - w_{FE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2}. \end{aligned} \quad (68)$$

Theorem 3 is proved. ■

Next, we bound the expected model gap at the BE, S, and FE during all clients' local training, i.e., $\mathbb{E}[\|w_{T, BE}^M - w_{BE}^*\|^2]$, $\mathbb{E}[\|w_{T, S}^M - w_S^*\|^2]$, and $\mathbb{E}[\|w_{T, FE}^M - w_{FE}^*\|^2]$, in Theorem 4. N is the number of all the clients (i.e., $N_R + N_{NR}$). As previously mentioned, M is the number of T -com rounds.

Theorem 4: Under Assumptions 1-4, the upper bounds on the expected deviations of BE, S, and FE models from the optimal models during the global model aggregation using all the clients are given by:

$$\begin{aligned} & \mathbb{E}[\|w_{T, BE}^M - w_{BE}^*\|^2] \\ & \leq \sum_{n=1}^{N_R} l_n \left(p_{R,n}(1 - \mu\eta_n)^{E_n K} \|w_{BE}^0 - w_{BE}^*\|^2 + \frac{2\eta_n^2 r^2 E_n K}{N^2} \right) \\ & \quad + \sum_{n=1}^{N_{NR}} l_n \left(p_{NR,n}(1 - \mu\eta_n)^{E_n} \|w_{BE}^0 - w_{BE}^*\|^2 + \frac{2\eta_n^2 r^2 E_n}{N^2} \right), \end{aligned} \quad (69)$$

$$\begin{aligned} & \mathbb{E}[\|w_{T, S}^M - w_S^*\|^2] \\ & \leq \sum_{n=1}^{N_R} l_n \left(p_{R,n}(1 - \mu\eta_n)^{E_n K} \|w_S^0 - w_S^*\|^2 + \frac{2\eta_n^2 r^2 E_n K}{N^2} \right) \\ & \quad + \sum_{n=1}^{N_{NR}} l_n \left(p_{NR,n}(1 - \mu\eta_n)^{E_n} \|w_S^0 - w_S^*\|^2 + \frac{2\eta_n^2 r^2 E_n}{N^2} \right), \end{aligned} \quad (70)$$

$$\begin{aligned} & \mathbb{E}[\|w_{T, FE}^M - w_{FE}^*\|^2] \\ & \leq \sum_{n=1}^{N_R} l_n \left(p_{R,n}(1 - \mu\eta_n)^{E_n K} \|w_{FE}^0 - w_{FE}^*\|^2 + \frac{2\eta_n^2 r^2 E_n K}{N^2} \right) \\ & \quad + \sum_{n=1}^{N_{NR}} l_n \left(p_{NR,n}(1 - \mu\eta_n)^{E_n} \|w_{FE}^0 - w_{FE}^*\|^2 + \frac{2\eta_n^2 r^2 E_n}{N^2} \right), \end{aligned} \quad (71)$$

where l_n is the local validation loss ratios of clients, that was defined under equation (4).

Proof: After each T -com round, the weights of all clients (both R and N_R) are aggregated based on l_n of clients.

$w_{R,BE}^K$ is the BE model created by the n^{th} R client after K^{th} R -com rounds. $w_{NR,BE}^{E_n}$ is the BE model created by the n^{th} NR client after E_n local epochs. We can express this aggregation as:

$$w_{T,BE}^M = \sum_{n=1}^{N_R} l_n w_{R,BE}^K + \sum_{n=1}^{N_{NR}} l_n w_{NR,BE}^{E_n}. \quad (72)$$

Similar in deriving (57) in Theorem 2, and in deriving (65) in Theorem 3, using the norm of a combination of vectors, we conclude:

$$\mathbb{E}[\|w_{T,BE}^M - w_{BE}^*\|^2] \leq \left[\sum_{n=1}^{N_R} l_n \mathbb{E}[\|w_{R,BE}^K - w_{BE}^*\|^2] \right]$$

$$+ \sum_{n=1}^{N_{NR}} l_n \mathbb{E}[\|w_{NR,BE}^{E_n} - w_{BE}^*\|^2]. \quad (73)$$

$$\mathbb{E}[\|w_{T,BE}^M - w_{BE}^*\|^2] \leq \left[\sum_{n=1}^{N_R} l_n C_2 + \sum_{n=1}^{N_{NR}} l_n C_3 \right]. \quad (74)$$

where, $C_2 = \mathbb{E}[\|w_{R,BE}^K - w_{BE}^*\|^2]$, and $C_3 = \mathbb{E}[\|w_{NR,BE}^{E_n} - w_{BE}^*\|^2]$.

Next, we bound C_2 and C_3 as follows:

$$C_2 = \mathbb{E}[\|w_{R,BE}^K - w_{BE}^*\|^2] \quad (75)$$

$$\begin{aligned} \mathbb{E}[F(w^M)] - F(w^*) &\leq \frac{L}{2N} \left[\sum_{n=1}^{N_R} l_n \left(p_{R,n}(1 - \mu\eta_n)^{E_n K} [\|w_{FE}^0 - w_{FE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n K}{N^2} \right) \right. \\ &\quad + \sum_{n=1}^{N_{NR}} l_n \left(p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_{FE}^0 - w_{FE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2} \right) \Big] \\ &\quad + \left[\sum_{n=1}^{N_R} l_n \left(p_{R,n}(1 - \mu\eta_n)^{E_n K} [\|w_S^0 - w_S^*\|^2] + \frac{2\eta_n^2 r^2 E_n K}{N^2} \right) \right. \\ &\quad + \sum_{n=1}^{N_{NR}} l_n \left(p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_S^0 - w_S^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2} \right) \Big] \\ &\quad + \left[\sum_{n=1}^{N_R} l_n \left(p_{R,n}(1 - \mu\eta_n)^{E_n K} [\|w_{BE}^0 - w_{BE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n K}{N^2} \right) \right. \\ &\quad + \sum_{n=1}^{N_{NR}} l_n \left(p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_{BE}^0 - w_{BE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2} \right) \Big]. \end{aligned}$$

$$\begin{aligned} \mathbb{E}[F(w^M)] - F(w^*) &\leq \frac{L}{2N} \left[\sum_{n=1}^{N_R} l_n \left(p_{R,n}(1 - \mu\eta_n)^{E_n K} [\|w_{FE}^0 - w_{FE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n K}{N^2} \right) \right. \\ &\quad + \sum_{n=1}^{N_{NR}} l_n \left(p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_{FE}^0 - w_{FE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2} \right) \Big] \\ &\quad + \left[\sum_{n=1}^{N_R} l_n \left(p_{R,n}(1 - \mu\eta_n)^{E_n K} [\|w_S^0 - w_S^*\|^2] + \frac{2\eta_n^2 r^2 E_n K}{N^2} \right) \right. \\ &\quad + \sum_{n=1}^{N_{NR}} l_n \left(p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_S^0 - w_S^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2} \right) \Big] \\ &\quad + \left[\sum_{n=1}^{N_R} l_n \left(p_{R,n}(1 - \mu\eta_n)^{E_n K} [\|w_{BE}^0 - w_{BE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n K}{N^2} \right) \right. \\ &\quad + \sum_{n=1}^{N_{NR}} l_n \left(p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_{BE}^0 - w_{BE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2} \right) \Big]. \end{aligned}$$

From (58) in Theorem 2,

$$C_2 \leq \sum_{n=1}^{N_R} p_{R,n}(1 - \mu\eta_n)^{E_n K} [\|w_{BE}^0 - w_{BE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n K}{N^2}. \quad (76)$$

$$C_3 = \mathbb{E}[\|w_{NR, BE}^{E_n} - w_{BE}^*\|^2] \quad (77)$$

From (66) in Theorem 3,

$$C_3 \leq \sum_{n=1}^{N_{NR}} p_{NR,n}(1 - \mu\eta_n)^{E_n} [\|w_{BE}^0 - w_{BE}^*\|^2] + \frac{2\eta_n^2 r^2 E_n}{N^2}. \quad (78)$$

Substituting (76) and (78) into (74):

$$\begin{aligned} & \mathbb{E}[\|w_{T, BE}^M - w_{BE}^*\|^2] \\ & \leq \sum_{n=1}^{N_R} l_n \left(p_{R,n}(1 - \mu\eta_n)^{E_n K} \|w_{BE}^0 - w_{BE}^*\|^2 + \frac{2\eta_n^2 r^2 E_n K}{N^2} \right) \\ & \quad + \sum_{n=1}^{N_{NR}} l_n \left(p_{NR,n}(1 - \mu\eta_n)^{E_n} \|w_{BE}^0 - w_{BE}^*\|^2 + \frac{2\eta_n^2 r^2 E_n}{N^2} \right). \end{aligned} \quad (79)$$

Similarly,

$$\begin{aligned} & \mathbb{E}[\|w_{T, S}^M - w_S^*\|^2] \\ & \leq \sum_{n=1}^{N_R} l_n \left(p_{R,n}(1 - \mu\eta_n)^{E_n K} \|w_S^0 - w_S^*\|^2 + \frac{2\eta_n^2 r^2 E_n K}{N^2} \right) \\ & \quad + \sum_{n=1}^{N_{NR}} l_n \left(p_{NR,n}(1 - \mu\eta_n)^{E_n} \|w_S^0 - w_S^*\|^2 + \frac{2\eta_n^2 r^2 E_n}{N^2} \right). \end{aligned} \quad (80)$$

and,

$$\begin{aligned} & \mathbb{E}[\|w_{T, FE}^M - w_{FE}^*\|^2] \\ & \leq \sum_{n=1}^{N_R} l_n \left(p_{R,n}(1 - \mu\eta_n)^{E_n K} \|w_{FE}^0 - w_{FE}^*\|^2 + \frac{2\eta_n^2 r^2 E_n K}{N^2} \right) \\ & \quad + \sum_{n=1}^{N_{NR}} l_n \left(p_{NR,n}(1 - \mu\eta_n)^{E_n} \|w_{FE}^0 - w_{FE}^*\|^2 + \frac{2\eta_n^2 r^2 E_n}{N^2} \right). \end{aligned} \quad (81)$$

Theorem 4 is proved. ■

Finally, we present an upper bound on the deviation of the expected loss of the AASFL-created global model from the optimal model, i.e., $\mathbb{E}[F(w^M)] - F(w^*)$.

Theorem 5: An upper bound on the deviation of the expected loss of the AASFL-created global model from the optimal model can be derived as shown at the bottom of previous page.

Proof: Telescoping and substituting with Lemma 1 (equation 19),

$$\begin{aligned} \mathbb{E}[F(w^M)] - F(w^*) & \leq \frac{L}{2} (\mathbb{E}[\|w_{T, FE}^M - w_{T, FE}^*\|^2] \\ & \quad + \mathbb{E}[\|w_{T, S}^M - w_{T, S}^*\|^2] \\ & \quad + \mathbb{E}[\|w_{T, BE}^M - w_{T, BE}^*\|^2]) \end{aligned} \quad (82)$$

Substituting (79), (80) and (81) into (82) we obtain the equations as shown at the bottom of previous page.

Theorem 5 is proved. ■

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